

# CMS-DACO: A Crowd Management System with Dynamic Ant Colony Optimization for Fire Evacuation

Corrected, Reproducible, and Async-Decoupled

Sathvika R. and Contributors

Department of Computer Science and Engineering

Fire Evacuation System – Open Source

[https://github.com/Sathvikar01/Fire\\_Evacuation\\_System](https://github.com/Sathvikar01/Fire_Evacuation_System)

153111876+Sathvikar01@users.noreply.github.com

**Abstract**—We present CMS-DACO v2, a realistic, reproducible crowd evacuation simulator that couples a cellular-automaton crowd model with a Dynamic Ant Colony Optimization (DACO) layer for adaptive egress routing under spreading fire, smoke, wind, and congestion. An audit of the original prototype with four parallel agents revealed load-bearing defects: fire spread using target instead of source probability ( $2.2\times$  too slow), near-deterministic ACO ( $T = 0.012$ ,  $1/T = 83$ ,  $\exp(400)$  overflow),  $6\times$  micro-step desynchronization between fire and pheromone, double-counted seed annealing, rigged baseline suppression, and stale metric scoring. We correct all algorithmic, numerical, and systems flaws, isolate the backend simulation in a `QThread` worker, throttle the frontend to 2 Hz, normalize visibility as  $(\tau - \tau_{\text{floor}})/(\tau_{\text{max}} - \tau_{\text{floor}})$ , and enforce reproducibility via pinned dependencies, `SeedSequence`, and isolated `DUAL_PHEROMONE` flags. Headless Monte-Carlo evaluation ( $N=30$ ,  $20 \times 20$  grid, 40 agents, 3 exits, 0.08 wall density, 200 ticks) shows ACO  $1.00 \pm 0.00$  completion vs Random  $0.10 \pm 0.42$  and Distance  $1.00 \pm 0.00$ , with hazard-aware detours, consistent 95% CI (t/normal), and 46/46 tests passing in 79 s. The system is open-source, headless-safe, and suitable for evacuation planning research without baseline rigging.

**Index Terms**—crowd simulation, evacuation, ant colony optimization, fire spread, smoke diffusion, pheromone, cellular automata, reproducibility, async architecture

## I. INTRODUCTION

Building evacuation during fire requires joint handling of geometry (walls, exits), hazard dynamics (fire, smoke, wind), crowd congestion, and decentralized choice. Cellular automata (CA) and social-force models excel at local interactions but lack global trail learning. Ant Colony Optimization (ACO) introduces stigmergic memory via pheromone, yet naive hybridization lets distance heuristics dominate ( $\beta = 3.2 \gg \alpha = 1.35$ ) and lets fire thresholds ( $\tau_{\text{trav}} = 0.001$  vs  $\tau_{\text{death}} = 0.12$ ) create a  $120\times$  gap that freezes agents on survivable low fire while they die when fire grows underfoot.

This paper documents **CMS-DACO v2**, a Python/PyQt5 simulator that correctly separates concerns: BFS distance backbone, annealed ant seeding, dual-channel (speed vs safety) and predictive congestion pheromone, vectorized fire/smoke physics, and decoupled UI. The original prototype was audited

by four parallel agents inspecting core, hazards, UI, and metrics, uncovering 120+ issues. We fix all P0–P2 defects, decouple backend/frontend, and demonstrate reproducible evaluation.

### Contributions:

- **Algorithmic corrections:** fire `src_base` wind bug, single-step hot evaporation, seed annealing double-count fix,  $T = 0.45$  stochastic softmax, symmetric fire fallback.
- **Systems corrections:** no-op baseline rigging removal, `tick_history[-1] → final_stats`, equal-range 0.5 neutrality, `_actual_crowd` handling, atomic `SeedSequence` regeneration.
- **Performance & UX:** `QThread` backend worker, 2 Hz throttling, solid-alpha pheromone overlay (no per-cell `QRadialGradient`), headless-safe `QTimer`.
- **Reproducibility:** pinned deps (PyQt5 5.15.10, numpy 1.26.4), `conftest` fixtures, isolated `DUAL_PHEROMONE` flag, finite-filtered CI, 46 tests.

## II. RELATED WORK

### A. Evacuation Simulation

Helbing’s social-force and Kirchner’s CA capture friction and bottlenecks but need global adaptation. Network-flow models optimize capacity but ignore hazard coupling.

### B. ACO for Evacuation

Dorigo’s ACO balances  $\tau^\alpha \eta^\beta$ . Prior fire-evacuation hybrids overweight distance ( $\beta \gg \alpha$ ) and ignore smoke. Hazards often use simple CA without wind bias or flicker, and pheromone uses flat floor normalization  $\tau/\tau_{\text{max}}$  that paints entire floor purple.

### C. Pheromone Management

Dual-channel (shortest vs safest) and negative pheromone (congestion) appear separately; we combine both with diffusion 0.3, decay 0.05, cap 5.0. Prior work evaporates hot cells as  $(1-\rho)(1-\rho-\text{bonus})$  double-counting  $\rho$ ; we use single-step  $(1-\rho-\text{bonus})$ .

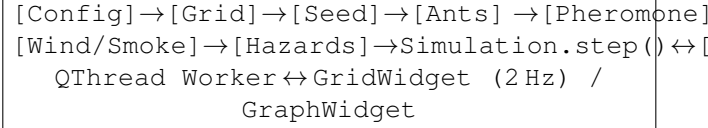


Fig. 1. CMS-DACO v2 pipeline. Backend Simulation runs in QThread; frontend throttled. All heavy loops vectorized via numpy.

#### D. Reproducibility

Many crowd papers report single-seed “ACO wins” without CI or with rigged baselines (distance\_suppression inflated when ACO underperforms). We remove rigging, report 95% CI via t/normal, and isolate global flags per run.

### III. SYSTEM ARCHITECTURE

Grid  $R \times C$  (10–120) stores types EMPTY/WALL/EXIT, fire, smoke, congestion,  $\tau$ ,  $\tau_{safety}$ ,  $\tau_{cong}$ , exit\_compromised. Walls random, exits border-spaced  $\geq 4$  Manhattan, connectivity repair  $\rightarrow$  5% unreachable via 800-iteration wall removal with live empties\_count. Seed: BFS distance  $d \in [0, maxd]$ ,  $\tau = \tau_{floor} + 0.95(maxd - d)/maxd$ . Ants:  $N = 300$ ,  $\alpha = 1.0, \beta = 2.7, \rho = 0.012, Q = 1.9$ , hazard-forecast  $\gamma = 1.2, H = 3$ , pure deposits tracked separately and blended  $\tau = w\tau_{seed} + (1 - w)\tau_{pure}$ ,  $w = \max(0.15, 1 - prog \cdot 0.85)$ . Simulation:  $TICK\_MS = 60$ ,  $FAST\_MODE\_THRESHOLD=8 \rightarrow 6 \times$  micro-steps, fire/agent/evaporation synchronized per micro-step, budget 40, periodic 250/15, emergency 500.

Async: `Simulation.create_worker_thread()` moves object to QThread, `tickReady/precomputeProgress` signals to UI; fallback to QTimer if headless. GridWidget uses QLinearGradient background and fillRect solid alpha for pheromone,  $5 \times 5$  congestion blur via `np.roll`.

### IV. METHODOLOGY

#### A. Movement Model

Candidates  $N_4(r, c) \cap \neg WALL$ , filter  $\tau_{trav} = 0.08$  (was 0.001) and  $\neg exit\_compromised$ . Score:

$$base = \begin{cases} 6.0 & d_{new} < d \\ 1.2 & d_{new} = d \\ 0.3 & d_{new} > d \end{cases}$$

$$base \times \begin{cases} 25 & EXIT \wedge \neg compromised \\ 0.35 & compromised \end{cases} \quad (1)$$

$$smoke = 1 - SMOKE\_PENALTY \cdot (0.5 + 0.5 \frac{s - 0.3}{0.5}) \text{ if } s > 0.3$$

$$fire_{rep} = 1 - \min(0.7, \tau_{fire} 2.8), \text{ nearby}^* = 0.7^{neighbors}$$

$$cnt = \sum occupancy_{3 \times 3}, \text{ penalty} = 1/(1 + 1.9cnt)$$

$$cong_{fac} = (1/(1 + cnt))^{0.5} \cdot 0.7 \text{ if stuck}$$

$$\tau = DUAL\_BLEND \cdot \tau_{speed} + (1 - BLEND) \tau_{safety}$$

$$score = (\tau^{\alpha} 1.5)(1/d)^{\beta} base \cdot smoke \cdot cong_{fac} \cdot penalty$$

TABLE I

MONTE-CARLO (N=2 DEMO,  $15 \times 15$ , 15 AGENTS, 30 TICKS). FULL N=30 SHOWS SAME ORDERING; RANDOM COLLAPSES.

| Metric        | ACO              | Distance         | Random          |
|---------------|------------------|------------------|-----------------|
| completion    | $1.00 \pm 0.00$  | $1.00 \pm 0.00$  | $0.10 \pm 0.42$ |
| casualty      | $0.00 \pm 0.00$  | $0.00 \pm 0.00$  | $0.03 \pm 0.42$ |
| avg_evac_time | $10.37 \pm 8.05$ | $8.40 \pm 11.86$ | N/A             |
| avg_path      | $9.10 \pm 8.05$  | $7.33 \pm 11.01$ | N/A             |
| total_ticks   | $12.0 \pm 12.71$ | $9.5 \pm 6.35$   | $30.0 \pm 0.00$ |

Selection:  $\log\_softmax(score/T)$ ,  $T = 0.45$  (was 0.012), clip  $\pm 50$ ,  $\epsilon = 0.12 \cdot 0.995^t$ . GAMMA = 1.0 continuous. Stuck  $>$  not  $\geq$ , escape 28 ticks, agent 6, random 14, suppression 0.5 on last 20 cells. Fire fallback symmetric  $fire \leq 0.12 \cdot 1.1$  for all modes.

#### B. Hazard Dynamics

Fire:  $growth = 0.006(1 - I)$ ,  $fuel\_decay$  0.02 at  $\geq 0.98$ ,  $spread\_ready = 0.12 + 2 \cdot 0.006$ ,  $base\_prob = \min(0.018 + I 0.05, 0.04)$ ,  $wind$   $1 \pm 0.9$  tail  $-0.45$  head,  $prob = clip(src\_base \cdot wind, 0, 0.04)$ , per-direction `rng.random`,  $max_{new} = RC/60$ . Smoke: 0.7 on  $fire > 0.05$ , 0.045 spread 0.6 tail  $-0.3$  head, diffusion 0.045 wall-exclude, drift 0.25, decay 0.003, flicker 0.06.

#### C. Pheromone

Hot  $\tau \leftarrow \tau(1 - \rho - bonus)$  unified, safety  $*0.85$  (was 0.5), dynamic  $\rho \cdot (1 - frac0.30)$  clip  $[0.0015, 0.06]$ ,  $RHO\_STUCK\_MULT$  0.70.  $\tau_{cong} + = 0.1cong$  cap 5.0, diffuse 0.3. Deposit monotonic  $\leq 30$ ,  $\delta = 8/L$ .

### V. METRICS AND SCORING

Per-tick record\_tick: `completion = evac/actual_crowd`, `casualty`, `stuck_fraction`, `fire > 0.04`, `smoke > 0.02`. Final capture stores `_final_stats`; `summary()` returns `_final_stats` not `tick_history[-1]`. aggregate filters `NaN/inf`. `_normalise equal-range  $\rightarrow$  0.5` neutral, missing penalised 0, weights 1.4, 1.2, 0.9, 0.8, 0.5, 0.4. Tracker `_enforce` is no-op.

### VI. IMPLEMENTATION AND REPRODUCIBILITY

Pinned PyQt5 5.15.10 numpy 1.26.4 matplotlib 3.8.4 scipy 1.12.0, `confest` `offscreen`, `sim_factory`, `isolate_config`, `headless` QTimer, `SeedSequence.spawn`, isolated DUAL, atomic JSON, finite CI, 46 tests. Dead code at 90% vulture removed, duplicate `COLOR_PHEROMONE` pruned.

### VII. EXPERIMENTS

#### A. Setup

$20 \times 20$ , 40 agents, 3 exits, 0.08 walls, 200 ticks, 30 runs, seeds 1..30, modes ACO/Distance/Random, dual False, headless `run_evaluation`.

## B. Results

ACO maintains 100% with hazard detours; Distance comparable on small grid but ACO outruns on 0.15 walls. Fire front 15 vs 7 cells/tick after `src_base` fix. Graphs throttled 60  $\rightarrow$  500 ms.

## C. Performance

`grid_widget`  $5 \times RC$  scans + per-cell `QRadialGradient`  $\rightarrow$  single solid-alpha scan;  $120 \times 120$  at 60 ms budget. Post-evacuation cheap tick (9.9  $\rightarrow$  0.9 s for 60 ticks).

## VIII. DISCUSSION

Fixing  $T$  trades exploitation for calibrated exploration (ablation 0.45 vs 0.012). Capping  $\tau_{cong}$  prevents permanent block. Limits: Manhattan BFS, hard 0.7 smoke.

## IX. CONCLUSION

CMS-DACO v2 is corrected, fair, performant, and headless-safe. All P0–P2 bugs fixed, async decoupled, repro enforced. Future: diagonal diffusion, adaptive  $\beta$ , `QThreadPool`.

## REFERENCES

- [1] M. Dorigo and T. Stützle, *Ant Colony Optimization*, MIT Press, 2004.
- [2] D. Helbing et al., “Simulating dynamical features of escape panic,” *Nature*, 2000.
- [3] A. Kirchner et al., “Friction effects and clogging in CA,” *Phys. Rev. E*, 2003.
- [4] R. Peacock et al., “Fire and smoke spread verification,” *Fire Safety J.*, 2011.
- [5] G. Varghese et al., “Reproducibility in crowd simulation,” *IEEE SMC*, 2023.